

Inputs

Image / Visual evidence

documents, charts, science diagrams, natural images

Question prompt

question text + optional answer choices

Target format

Answer only: "Answer: <answer>"
Rationale target: "Reasoning: <rationale> Answer: <answer>"

Qwen2-VL Branch

Qwen2-VL 2B / 7B

Visual encoder
frozen

Language decoder
4-bit NF4 frozen base

QLoRA adapters
trainable
attention + MLP layers

BLIP-2 Branch

BLIP-2 OPT-2.7B

ViT vision encoder
frozen

Q-Former
trainable in BLIP runs
32 learned query tokens

OPT language model
frozen

Qwen2-VL Supervision

Answer-only CE

train only answer tokens

Rationale-generative CE

train full rationale+answer

Explanation-aware CE

separate rationale span
and answer span

Loss

$L = \alpha * CE_{\text{answer}} + (1-\alpha) * CE_{\text{rationale}}$

BLIP-2 Supervision

Generative CE

rationale + answer targets

Contrastive-enhanced

generative CE + InfoNCE image-answer loss

Image embedding

pooled Q-Former queries

Text embedding

pooled answer tokens projected to Q-Former dim

Model Outputs

Predicted answer
Generated rationale when required
Image-answer embeddings for BLIP-2 retrieval check

Evaluations

Correctness: MC accuracy, ANLS, relaxed accuracy, VQA accuracy
Rationale quality: ROUGE-L, BLEU on rationale datasets
Grounding: image masking drift / evidence sensitivity
Transfer: source adapter evaluated on other domains